ROCSCIENCE

Case12_Water_Surcharge
SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

Case12_Water_Surcharge

Project Summary

File Name:	Case12_Water_Surcharge.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:02.879s
Project Title:	SLIDE - An Interactive Slope Stability Program
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.0001
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Loading

1 Distributed Load present

Distributed Load 1	
Distribution:	Constant
Magnitude [kPa]:	100
Orientation:	Normal to boundary

Materials

top layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	29.4 kPa
Phi	12 °
Water Surface	Water Table
Hu Type	Automatically Calculated
Specify alternate strength type above water surface	No

middle layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	9.8 kPa
Phi	5 °
Water Surface	Water Table
Hu Type	Automatically Calculated
Specify alternate strength type above water surface	No

lower layer	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	294 kPa
Phi	40 °
Water Surface	Water Table
Hu Type	Automatically Calculated
Specify alternate strength type above water surface	No

Global Minimums

Method: bishop simplified

FS	0.310792
Center:	27.046, 45.140
Radius:	31.430
Left Slip Surface Endpoint:	18.041, 15.027
Right Slip Surface Endpoint:	56.796, 35.000
Left Slope Intercept:	18.041 19.000
Right Slope Intercept:	56.796 35.000
Resisting Moment:	30004 kN-m
Driving Moment:	96540.5 kN-m
Total Slice Area:	363.1 m2
Surface Horizontal Width:	38.7552 m
Surface Average Height:	9.36907 m

Method: spencer

FS	0.313910
Center:	27.046, 45.140
Radius:	31.430
Left Slip Surface Endpoint:	18.041, 15.027
Right Slip Surface Endpoint:	56.796, 35.000
Left Slope Intercept:	18.041 19.000
Right Slope Intercept:	56.796 35.000
Resisting Moment:	30305 kN-m
Driving Moment:	96540.5 kN-m
Resisting Horizontal Force:	764.992 kN
Driving Horizontal Force:	2436.98 kN
Total Slice Area:	363.1 m2
Surface Horizontal Width:	38.7552 m
Surface Average Height:	9.36907 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	10032
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	5426
Number of Invalid Surfaces:	4606

Error Codes

Error Code -108 reported for 285 surfaces
Error Code -111 reported for 4319 surfaces
Error Code -112 reported for 2 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge
-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.310792

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.765317	33.1583	-15.9247	middle layer	9.8	5	35.2908	10.9681	53.3955	40.0439	13.3516	43.3262	3.28227
2	0.765317	39.6723	-14.4788	middle layer	9.8	5	36.9649	11.4884	61.3828	42.0843	19.2985	51.8376	9.75329
3	0.765317	45.8903	-13.0422	middle layer	9.8	5	38.562	11.9848	68.8952	43.9232	24.972	59.9625	16.0393
4	0.765317	51.818	-11.614	middle layer	9.8	5	40.0849	12.4581	75.9463	45.5643	30.382	67.7078	22.1435
5	0.765317	57.4599	-10.1931	middle layer	9.8	5	41.5362	12.9091	82.5481	47.0107	35.5374	75.0798	28.0691
6	0.765317	62.8202	-8.77844	middle layer	9.8	5	42.918	13.3386	88.7113	48.2654	40.4459	82.0838	33.8184
7	0.765317	67.9023	-7.36918	middle layer	9.8	5	44.2321	13.747	94.4449	49.3306	45.1143	88.7244	39.3938
8	0.765317	72.7963	-5.9644	middle layer	9.8	5	45.5134	14.1452	99.8742	50.2083	49.6659	95.1192	44.9109
9	0.765317	79.9759	-4.56321	middle layer	9.8	5	49.1969	15.29	108.427	45.6757	62.7511	104.5	58.8246
10	0.765317	88.0693	-3.16475	middle layer	9.8	5	51.0083	15.853	117.896	48.71	69.1858	115.075	66.3655
11	0.765317	95.8929	-1.76817	middle layer	9.8	5	52.7407	16.3914	126.926	51.5862	75.34	125.298	73.7119
12	0.765317	103.448	-0.372651	middle layer	9.8	5	54.3956	16.9057	135.524	54.3047	81.2192	135.17	80.8655
13	0.765317	110.734	1.02265	middle layer	9.8	5	55.9738	17.3962	143.691	56.8657	86.8257	144.691	87.8248
14	0.765317	117.751	2.41856	middle layer	9.8	5	57.4764	17.8632	151.432	59.2692	92.163	153.86	94.5906
15	0.765317	124.5	3.81591	middle layer	9.8	5	58.9038	18.3068	158.749	61.5148	97.2338	162.677	101.163
16	0.765317	130.978	5.21553	middle layer	9.8	5	60.2567	18.7273	165.641	63.6019	102.039	171.141	107.539
17	0.765317	137.184	6.61829	middle layer	9.8	5	61.5353	19.1247	172.111	65.5296	106.581	179.251	113.721
18	0.765317	143.116	8.02504	middle layer	9.8	5	62.7397	19.499	178.157	67.2966	110.86	187.002	119.706
19	0.765317	148.771	9.43669	middle layer	9.8	5	63.87	19.8503	183.777	68.9015	114.875	194.392	125.491
20	0.765317	154.147	10.8541	middle layer	9.8	5	64.9257	20.1784	188.968	70.3423	118.626	201.417	131.075
21	0.765317	159.24	12.2784	middle layer	9.8	5	65.9065	20.4832	193.727	71.6169	122.11	208.07	136.453
22	0.765317	164.044	13.7103	middle layer	9.8	5	66.8119	20.7646	198.048	72.7227	125.325	214.348	141.625
23	0.765317	168.556	15.1511	middle layer	9.8	5	67.6411	21.0223	201.928	73.6568	128.271	220.243	146.587
24	0.765317	172.769	16.6017	middle layer	9.8	5	68.3927	21.2559	205.357	74.4158	130.942	225.748	151.333
25	1.15033	266.889	18.4349	middle layer	9.8	5	69.2128	21.5108	208.94	75.0841	133.856	232.011	156.927
26	0.783875	186.411	20.3021	middle layer	9.8	5	69.9329	21.7346	211.935	75.5227	136.413	237.807	162.284
27	0.783875	189.665	21.8336	middle layer	9.8	5	70.412	21.8835	213.748	75.6325	138.115	241.958	166.326
28	0.783875	192.558	23.3818	middle layer	9.8	5	70.8023	22.0048	215.036	75.5357	139.501	245.648	170.113
29	0.783875	195.077	24.9482	middle layer	9.8	5	71.1006	22.0975	215.786	75.2251	140.561	248.863	173.637
30	0.783875	197.21	26.5349	middle layer	9.8	5	71.3043	22.1608	215.978	74.6929	141.285	251.583	176.89

31	0.783875	198.939	28.1439	middle layer	9.8	5	71.4098	22.1936	215.59	73.9298	141.66	253.79	179.86
32	0.783875	200.247	29.7774	middle layer	9.8	5	71.413	22.1946	214.597	72.9256	141.671	255.458	182.532
33	0.783875	201.113	31.438	middle layer	9.8	5	71.3091	22.1623	212.971	71.6686	141.302	256.563	184.894
34	0.783875	201.515	33.1287	middle layer	9.8	5	71.0929	22.0951	210.68	70.1454	140.535	257.076	186.93
35	0.783875	201.424	34.8526	middle layer	9.8	5	70.7583	21.9911	207.685	68.3405	139.344	256.959	188.619
36	0.783875	200.811	36.6135	middle layer	9.8	5	70.2975	21.8479	203.944	66.2364	137.708	256.178	189.941
37	0.783875	199.639	38.4155	middle layer	9.8	5	69.7022	21.6629	199.406	63.8123	135.594	254.682	190.87
38	0.783875	197.865	40.2638	middle layer	9.8	5	68.9625	21.433	194.01	61.0441	132.966	252.42	191.376
39	0.783875	193.659	42.1642	middle layer	9.8	5	82.8097	25.7366	240.06	57.9032	182.157	315.053	257.15
40	0.783875	183.21	44.1235	middle layer	9.8	5	86.5463	26.8979	249.785	54.3556	195.43	333.723	279.368
41	0.783875	171.583	46.1503	middle layer	9.8	5	82.847	25.7482	232.649	50.3598	182.289	318.891	268.531
42	0.783875	159.084	48.2548	middle layer	9.8	5	78.9853	24.548	214.435	45.8651	168.57	302.946	257.081
43	0.783875	145.603	50.4502	middle layer	9.8	5	74.9382	23.2902	195.002	40.8077	154.195	285.749	244.941
44	0.783875	130.997	52.7529	middle layer	9.8	5	70.6765	21.9657	174.161	35.1062	139.055	267.115	232.009
45	0.783875	115.078	55.185	middle layer	9.8	5	66.161	20.5623	151.666	28.653	123.013	246.806	218.153
46	0.783875	97.5895	57.7766	middle layer	9.8	5	61.3379	19.0633	127.182	21.3016	105.881	224.497	203.195
47	0.702152	71.0273	60.4129	top layer	29.4	12	101.174	31.444	22.9657	13.3494	9.61627	201.158	187.808
48	0.702152	53.7018	63.1252	top layer	29.4	12	72.6809	22.5886	-27.3857	4.65925	-32.0449	116.032	111.373
49	0.685792	33.5304	66.0832	top layer	29.4	12	50.3658	15.6533	-64.673	0	-64.673	48.8939	48.8939
50	0.685792	11.7756	69.4023	top layer	29.4	12	37.7127	11.7208	-83.174	0	-83.174	17.1713	17.1713

Global Minimum Query (spencer) - Safety Factor: 0.31391

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.765317	33.1583	-15.9247	middle layer	9.8	5	40.4596	12.7007	73.1986	40.0439	33.1547	61.6545	21.6106
2	0.765317	39.6723	-14.4788	middle layer	9.8	5	41.9504	13.1686	80.5881	42.0843	38.5038	69.7556	27.6713
3	0.765317	45.8903	-13.0422	middle layer	9.8	5	43.3302	13.6018	87.3776	43.9232	43.4544	77.3404	33.4172
4	0.765317	51.818	-11.614	middle layer	9.8	5	44.6065	14.0024	93.5982	45.5643	48.0339	84.4305	38.8662
5	0.765317	57.4599	-10.1931	middle layer	9.8	5	45.7861	14.3727	99.2769	47.0107	52.2662	91.0444	44.0337
6	0.765317	62.8202	-8.77844	middle layer	9.8	5	46.8746	14.7144	104.437	48.2654	56.172	97.1989	48.9335
7	0.765317	67.9023	-7.36918	middle layer	9.8	5	47.8773	15.0292	109.1	49.3306	59.7695	102.908	53.5775
8	0.765317	72.7963	-5.9644	middle layer	9.8	5	48.8395	15.3312	113.43	50.2083	63.2219	108.328	58.1194
9	0.765317	79.9759	-4.56321	middle layer	9.8	5	52.4932	16.4781	122.007	45.6757	76.3316	117.818	72.142
10	0.765317	88.0693	-3.16475	middle layer	9.8	5	54.2175	17.0194	131.229	48.71	82.5187	128.231	79.521
11	0.765317	95.8929	-1.76817	middle layer	9.8	5	55.8309	17.5259	139.894	51.5862	88.3074	138.17	86.5839
12	0.765317	103.448	-0.372651	middle layer	9.8	5	57.3378	17.9989	148.019	54.3047	93.7145	147.646	93.3415
13	0.765317	110.734	1.02265	middle layer	9.8	5	58.7418	18.4397	155.618	56.8657	98.7522	156.666	99.8007
14	0.765317	117.751	2.41856	middle layer	9.8	5	60.0465	18.8492	162.702	59.2692	103.433	165.239	105.969
15	0.765317	124.5	3.81591	middle layer	9.8	5	61.2549	19.2285	169.283	61.5148	107.768	173.369	111.854
16	0.765317	130.978	5.21553	middle layer	9.8	5	62.3693	19.5784	175.369	63.6019	111.767	181.062	117.46
17	0.765317	137.184	6.61829	middle layer	9.8	5	63.3924	19.8995	180.967	65.5296	115.437	188.322	122.792
18	0.765317	143.116	8.02504	middle layer	9.8	5	64.3258	20.1925	186.083	67.2966	118.786	195.152	127.856
19	0.765317	148.771	9.43669	middle layer	9.8	5	65.1712	20.4579	190.722	68.9015	121.82	201.554	132.652
20	0.765317	154.147	10.8541	middle layer	9.8	5	65.93	20.6961	194.885	70.3423	124.543	207.527	137.184
21	0.765317	159.24	12.2784	middle layer	9.8	5	66.6038	20.9076	198.577	71.6169	126.96	213.073	141.456
22	0.765317	164.044	13.7103	middle layer	9.8	5	67.1931	21.0926	201.798	72.7227	129.075	218.191	145.468
23	0.765317	168.556	15.1511	middle layer	9.8	5	67.6984	21.2512	204.545	73.6568	130.888	222.876	149.219
24	0.765317	172.769	16.6017	middle layer	9.8	5	68.1205	21.3837	206.819	74.4158	132.403	227.128	152.713
25	1.15033	266.889	18.4349	middle layer	9.8	5	68.5222	21.5098	208.927	75.0841	133.843	231.768	156.684
26	0.783875	186.411	20.3021	middle layer	9.8	5	68.8143	21.6015	210.415	75.5227	134.892	235.873	160.35
27	0.783875	189.665	21.8336	middle layer	9.8	5	68.9456	21.6427	210.996	75.6325	135.363	238.619	162.986
28	0.783875	192.558	23.3818	middle layer	9.8	5	68.9886	21.6562	211.053	75.5357	135.517	240.881	165.345
29	0.783875	195.077	24.9482	middle layer	9.8	5	68.9424	21.6417	210.576	75.2251	135.351	242.649	167.424
30	0.783875	197.21	26.5349	middle layer	9.8	5	68.8057	21.5988	209.553	74.6929	134.86	243.911	169.218
31	0.783875	198.939	28.1439	middle layer	9.8	5	68.5767	21.5269	207.969	73.9298	134.039	244.653	170.723
32	0.783875	200.247	29.7774	middle layer	9.8	5	68.2536	21.4255	205.806	72.9256	132.881	244.86	171.934
33	0.783875	201.113	31.438	middle layer	9.8	5	67.8344	21.2939	203.044	71.6686	131.375	244.512	172.844

34	0.783875	201.515	33.1287	middle layer	9.8	5	67.3158	21.1311	199.661	70.1454	129.516	243.592	173.446
35	0.783875	201.424	34.8526	middle layer	9.8	5	66.6949	20.9362	195.628	68.3405	127.287	242.073	173.732
36	0.783875	200.811	36.6135	middle layer	9.8	5	65.968	20.708	190.915	66.2364	124.679	239.931	173.695
37	0.783875	199.639	38.4155	middle layer	9.8	5	65.1301	20.445	185.485	63.8123	121.673	237.135	173.323
38	0.783875	197.865	40.2638	middle layer	9.8	5	64.176	20.1455	179.294	61.0441	118.25	233.65	172.606
39	0.783875	193.659	42.1642	middle layer	9.8	5	76.3107	23.9547	219.692	57.9032	161.789	288.799	230.896
40	0.783875	183.21	44.1235	middle layer	9.8	5	79.2644	24.8819	226.743	54.3556	172.387	303.619	249.263
41	0.783875	171.583	46.1503	middle layer	9.8	5	75.6143	23.7361	209.65	50.3598	159.29	288.363	238.003
42	0.783875	159.084	48.2548	middle layer	9.8	5	71.8868	22.566	191.781	45.8651	145.916	272.337	226.472
43	0.783875	145.603	50.4502	middle layer	9.8	5	68.0733	21.3689	173.041	40.8077	132.233	255.474	214.666
44	0.783875	130.997	52.7529	middle layer	9.8	5	64.1639	20.1417	153.312	35.1062	118.206	237.701	202.594
45	0.783875	115.078	55.185	middle layer	9.8	5	60.1456	18.8803	132.442	28.653	103.789	218.931	190.278
46	0.783875	97.5895	57.7766	middle layer	9.8	5	56.0025	17.5798	110.225	21.3016	88.923	199.075	177.773
47	0.702152	71.0273	60.4129	top layer	29.4	12	104.228	32.7181	28.9596	13.3494	15.6102	212.53	199.18
48	0.702152	53.7018	63.1252	top layer	29.4	12	79.3091	24.8959	-16.5308	4.65925	-21.19	139.967	135.307
49	0.685792	33.5304	66.0832	top layer	29.4	12	60.0261	18.8428	-49.6679	0	-49.6679	85.6815	85.6815
50	0.685792	11.7756	69.4023	top layer	29.4	12	49.1807	15.4383	-65.6847	0	-65.6847	65.1744	65.1744

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.310792

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	18.0408	15.0272	77.4154	0	0
2	18.8062	14.8089	57.2757	0	0
3	19.5715	14.6112	113.75	0	0
4	20.3368	14.434	168.976	0	0
5	21.1021	14.2767	222.546	0	0
6	21.8674	14.1391	274.086	0	0
7	22.6327	14.0209	323.255	0	0
8	23.3981	13.9219	369.74	0	0
9	24.1634	13.8419	413.347	0	0
10	24.9287	13.7809	457.621	0	0
11	25.694	13.7385	501.648	0	0
12	26.4593	13.7149	545.009	0	0
13	27.2246	13.7099	587.314	0	0
14	27.99	13.7236	628.188	0	0
15	28.7553	13.7559	667.281	0	0
16	29.5206	13.807	704.257	0	0
17	30.2859	13.8768	738.801	0	0
18	31.0512	13.9656	770.612	0	0
19	31.8165	14.0735	799.404	0	0
20	32.5819	14.2007	824.908	0	0
21	33.3472	14.3475	846.867	0	0
22	34.1125	14.514	865.038	0	0
23	34.8778	14.7007	879.193	0	0
24	35.6431	14.908	889.114	0	0
25	36.4084	15.1361	894.598	0	0
26	37.5588	15.5196	894.099	0	0
27	38.3427	15.8096	887.457	0	0
28	39.1265	16.1236	875.521	0	0
29	39.9104	16.4626	858.141	0	0
30	40.6943	16.8272	835.185	0	0
31	41.4782	17.2187	806.54	0	0
32	42.262	17.638	772.114	0	0
33	43.0459	18.0865	731.842	0	0
34	43.8298	18.5657	685.685	0	0
35	44.6137	19.0773	633.637	0	0
36	45.3975	19.6231	575.733	0	0
37	46.1814	20.2056	512.051	0	0
38	46.9653	20.8272	442.73	0	0
39	47.7491	21.4911	367.98	0	0
40	48.533	22.201	262.478	0	0
41	49.3169	22.9613	140.42	0	0
42	50.1008	23.7773	15.5202	0	0
43	50.8846	24.6557	-110.927	0	0
44	51.6685	25.6049	-237.288	0	0
45	52.4524	26.6359	-361.438	0	0
46	53.2363	27.7631	-480.538	0	0
47	54.0201	29.0067	-590.626	0	0
48	54.7223	30.2434	-547.988	0	0
49	55.4245	31.6289	-459.011	0	0
50	56.1102	33.1753	-324.464	0	0
51	56.796	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 0.31391

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	18.0408	15.0272	77.4154	0	0
2	18.8062	14.8089	65.5568	14.0304	12.0801
3	19.5715	14.6112	129.643	27.746	12.0801
4	20.3368	14.434	191.796	41.048	12.0801
5	21.1021	14.2767	251.604	53.848	12.0801
6	21.8674	14.1391	308.7	66.0676	12.0801
7	22.6327	14.0209	362.757	77.6368	12.0801
8	23.3981	13.9219	413.483	88.4932	12.0801
9	24.1634	13.8419	460.722	98.6031	12.0801
10	24.9287	13.7809	508.349	108.796	12.0801
11	25.694	13.7385	555.397	118.865	12.0801
12	26.4593	13.7149	601.432	128.718	12.0802
13	27.2246	13.7099	646.052	138.267	12.0801
14	27.99	13.7236	688.883	147.434	12.0801
15	28.7553	13.7559	729.58	156.144	12.0801
16	29.5206	13.807	767.82	164.328	12.0801
17	30.2859	13.8768	803.303	171.922	12.0801
18	31.0512	13.9656	835.75	178.866	12.0801
19	31.8165	14.0735	864.903	185.105	12.0801
20	32.5819	14.2007	890.521	190.588	12.0801
21	33.3472	14.3475	912.382	195.267	12.0801
22	34.1125	14.514	930.281	199.098	12.0802
23	34.8778	14.7007	944.029	202.04	12.0801
24	35.6431	14.908	953.454	204.057	12.0801
25	36.4084	15.1361	958.398	205.115	12.0801
26	37.5588	15.5196	957.112	204.84	12.0801
27	38.3427	15.8096	950.036	203.325	12.0801
28	39.1265	16.1236	937.817	200.71	12.0801
29	39.9104	16.4626	920.367	196.976	12.0801
30	40.6943	16.8272	897.621	192.108	12.0801
31	41.4782	17.2187	869.534	186.097	12.0802
32	42.262	17.638	836.086	178.938	12.0801
33	43.0459	18.0865	797.282	170.633	12.0801
34	43.8298	18.5657	753.16	161.19	12.0801
35	44.6137	19.0773	703.79	150.624	12.0801
36	45.3975	19.6231	649.284	138.959	12.0801
37	46.1814	20.2056	589.799	126.228	12.0801
38	46.9653	20.8272	525.549	112.477	12.0801
39	47.7491	21.4911	456.819	97.7679	12.0801
40	48.533	22.201	360.684	77.1931	12.0801
41	49.3169	22.9613	250.438	53.5984	12.0801
42	50.1008	23.7773	138.637	29.671	12.0802
43	50.8846	24.6557	26.5275	5.67738	12.0801
44	51.6685	25.6049	-84.3659	-18.0559	12.0801
45	52.4524	26.6359	-192.126	-41.1185	12.0801
46	53.2363	27.7631	-294.268	-62.9788	12.0801
47	54.0201	29.0067	-387.447	-82.921	12.0801
48	54.7223	30.2434	-350.075	-74.9226	12.0801
49	55.4245	31.6289	-271.482	-58.1023	12.0801
50	56.1102	33.1753	-153.512	-32.8544	12.0801
51	56.796	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 3 96, 3 96, 35 72, 35 48, 35 24, 19 18, 15 0, 15
Material Boundary	24, 19 72, 35
Material Boundary	0, 3 96, 35

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Water Table	0, 19 24, 19 56.976, 32.2524 96, 32.2524	Assigned to: top layer middle layer lower layer
Distributed Load	55, 35 48, 35	Constant DistributionOrientation: Normal to boundaryMagnitude: 100 kN/m2Creates Excess Pore Pressure: No